

Homework 5--Problem

Deadline: March 24, 2022

Problem 4.1

4.1 Calculate the difference in molar entropy (a) between liquid water and ice at -5°C , (b) between liquid water and its vapour at 95°C and 1.00 atm . The differences in heat capacities on melting and on vaporization are $37.3\text{ J K}^{-1}\text{ mol}^{-1}$ and $-41.9\text{ J K}^{-1}\text{ mol}^{-1}$, respectively. Distinguish between the entropy changes of the sample, the surroundings, and the total system, and discuss the spontaneity of the transitions at the two temperatures.

Problem 4.4

4.4 Consider a perfect gas contained in a cylinder and separated by a frictionless adiabatic piston into two sections A and B. All changes in B are isothermal; that is, a thermostat surrounds B to keep its temperature constant. There is 2.00 mol of the gas in each section. Initially, $T_A = T_B = 300\text{ K}$, $V_A = V_B = 2.00\text{ L}$. Heat is added to Section A and the piston moves to the right reversibly until the final volume of Section B is 1.00 L . Calculate (a) ΔS_A and ΔS_B , (b) ΔA_A and ΔA_B , (c) ΔG_A and ΔG_B , (d) ΔS of the total system and its surroundings. If numerical values cannot be obtained, indicate whether the values should be positive, negative, or zero or are indeterminate from the information given. (Assume $C_{V,m} = 20\text{ J K}^{-1}\text{ mol}^{-1}$.)

Problem 4.8

4.8 A sample of 1.00 mol of a monatomic perfect gas with $C_{V,m} = \frac{3}{2}R$, initially at 298 K and 10 L , is expanded, with the surroundings maintained at 298 K , to a final volume of 20 L , in three ways: (a) isothermally and reversibly, (b) isothermally against a constant external pressure of 0.50 atm , (c) adiabatically against a constant external pressure of 0.50 atm . Calculate ΔS , ΔS_{sur} , ΔH , ΔT , ΔA , and ΔG for each path where the data permit.

Problem 4.15

4.15 From standard enthalpies of formation, standard entropies, and standard heat capacities available from tables in the *Data section*, calculate: (a) the standard enthalpies and entropies at 298 K and 398 K for the reaction $\text{CO}_2(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g})$. Assume that the heat capacities are constant over the temperature range involved.

Problem 4.32

4.32 The cycle involved in the operation of an internal combustion engine is called the *Otto cycle*. Air can be considered to be the working substance and can be assumed to be a perfect gas. The cycle consists of the following steps: (1) reversible adiabatic compression from A to B, (2) reversible constant-volume pressure increase from B to C due to the combustion of a small amount of fuel, (3) reversible adiabatic expansion from C to D, and (4) reversible and constant-volume pressure decrease back to state A. Determine the change in entropy (of the system and of the surroundings) for each step of the cycle and determine an expression for the efficiency of the cycle, assuming that the heat is supplied in Step 2. Evaluate the efficiency for a compression ratio of 10:1. Assume that in state A, $V = 4.00 \text{ L}$, $p = 1.00 \text{ atm}$, and $T = 300 \text{ K}$, that $V_A = 10V_B$, $p_C/p_B = 5$, and that $C_{p,m} = \frac{7}{2}R$.